

Physics

Overview

Dust Creation

Astration

Gas-Dust Drag

Thermal Sputtering

Shock Destruction

Grain Growth

Subgrid Clumping

Coagulation

Shattering

Radiation Pressure

Physics Ladder

Table of contents

Execution Flow

Initial Grain Properties

Parameters

Dust Creation

At a Glance

Purpose	Primary Sources	Produces
Initialize the CosmicGrain dust lifecycle by injecting explicit dust superparticles from stellar sources.	Core-collapse supernovae (SNII) Asymptotic Giant Branch (AGB) stars	Explicit PartType6 dust superparticles.
Physics Ladder	Runtime Cost	Status
S0 — Dust Creation	Low	Implemented

Dust creation initializes the CosmicGrain dust lifecycle by injecting explicit dust superparticles into the simulation from stellar sources. These particles subsequently evolve through transport, growth, and destruction, allowing the complete lifecycle of cosmic dust to be followed self-consistently alongside gas, stars, and dark matter.

Scientific Motivation

Interstellar dust is believed to originate primarily from stellar ejecta. CosmicGrain currently models two stellar dust production channels: - Core-collapse supernovae (SNII) - Asymptotic Giant Branch (AGB) stars These sources dominate dust production during different epochs of cosmic evolution.

Physical Model

For each stellar feedback event, a fraction of the newly produced metals is assumed to condense into dust:

$$M_{\text{dust}} = f_{\text{cond}} M_{\text{metals}}$$

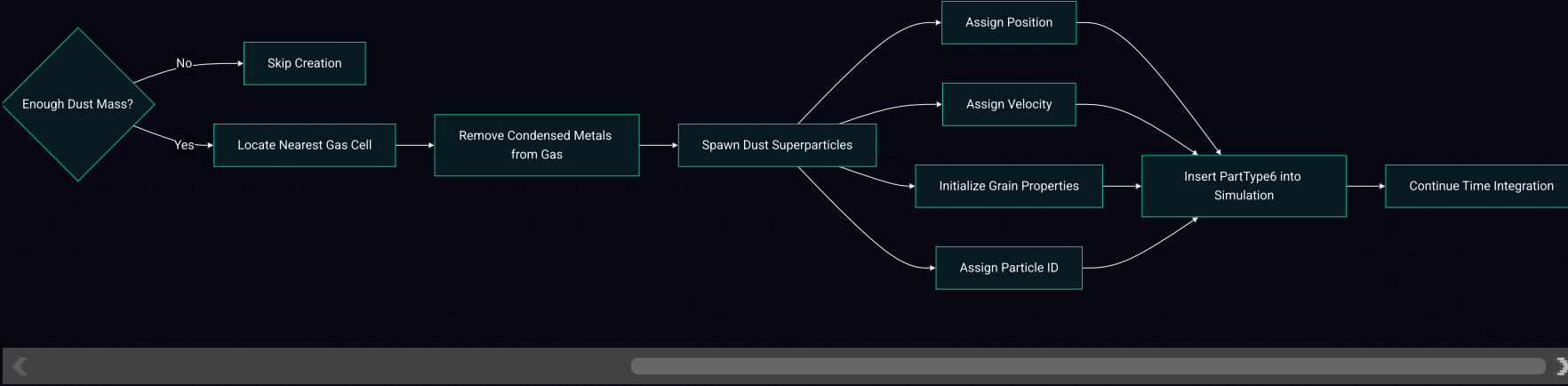
Separate condensation efficiencies are used for SNII and AGB stars.

Key Assumptions

- Dust forms immediately following stellar feedback. - Condensation efficiencies are fixed for each source. - Dust is represented by superparticles, not individual grains. - Initial grain properties depend on the stellar production channel.

How CosmicGrain Implements This

When stellar feedback occurs, CosmicGrain computes the metal mass ejected by the stellar particle and converts a fraction of those metals into dust. Rather than storing dust as a passive scalar on nearby gas cells, the code creates new **PartType6** particles. The surrounding gas cell is updated so that metals condensed into dust are removed from the gas phase.



Initial Grain Properties

Quantity	SNII	AGB
Grain radius	10 nm	100 nm
Carbon fraction	0.1	0.6
Initial velocity	100 km s ⁻¹	10 km s ⁻¹

Parameters

Parameter	Description
DustYieldSNII	SNII condensation efficiency
DustYieldAGB	AGB condensation efficiency
DustVelocitySNII	Initial SNII velocity kick
DustVelocityAGB	Initial AGB velocity kick
DustParticlesPerSNII	Number of particles spawned per SNII event
DustParticlesPerAGB	Number of particles spawned per AGB event

Where in the Code?

File or Function	Purpose		--
`dust.cc`	Dust physics implementation		
`create\_dust\_particles\_from\_feedback()`	Converts stellar ejecta into dust particles		
`spawn\_dust\_particle()`	Initializes new PartType6 particles		
`feedback.cc`	Stellar feedback interface		

Why This Matters

Dust creation establishes the initial conditions for the entire CosmicGrain dust lifecycle. Every later process—drag, growth, coagulation, shattering, sputtering, shock destruction, and radiation pressure—acts only on dust that first enters the simulation through stellar production.

Primary References

- Todini & Ferrara (2001) - Nozawa et al. (2003) - Ferrarotti & Gail (2006) - Dwek (1998) - McKinnon et al. (2016)